Name:- Syed Jasim Us Sani

Roll no:- B-51

Branch:- AI&DS(B)

Batch:- B3

Experiment:-06

AIM:- Implementing Web Scraping in Python with BeautifulSoup

Theory :-

1. What is web scraping? Role of Requests library Web scraping is the process of extracting data from websites automatically using programs.

Role of Requests library:

- Sends HTTP requests to web pages.
- Retrieves the HTML content of a webpage.
- Acts as the first step in scraping (getting the page data)

Example:-

```
import requests
url = "https://example.com"
response = requests.get(url)
print(response.text)
```

```
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial
```

2. What is BeautifulSoup? How does it help?

BeautifulSoup is a Python library used to parse HTML and XML documents.

How it helps:

- Converts raw HTML into a structured format (parse tree).
- Makes it easy to search and extract elements.
- Handles poorly formatted HTML.

Example:-

```
from bs4 import BeautifulSoup
html = "<html><body><h1>Hello</h1></body></html>"
soup = BeautifulSoup(html, "html.parser")
print(soup.h1.text)
```

```
Hello
```

3. Difference between Web Scraping and Web Crawling

Feature	Web Scraping	Web Crawling
Purpose	Extract specific data from web pages.	Discover and index web pages.
Goal	Obtain structured data for analysis.	Build a comprehensive index of the web.
Mechanism	Uses parsers (e.g., BeautifulSoup) to find and extract data from specific URLs.	Follows links to navigate and discover new web pages.
Output	Structured data (e.g., CSV, JSON, database).	A list of URLs and their content for indexing.
Scope	Focused on specific data points within a page or a set of pages.	Broad, aiming to cover a large part of the internet or a specific domain.
Interaction	Often interacts with a specific page once or few times.	Continuously explores and re-explores pages.
Example Tool	BeautifulSoup, Scrapy (often combined with Requests)	Googlebot, Bingbot, Scrapy (for comprehensive site mapping)

4. Commonly used methods in BeautifulSoup

Some important methods:

- `find()` → Finds first matching element.
- `find_all()` → Finds all matching elements.
- `get_text()` → Extracts text.
- `get()` → Gets attribute value.
- `select()` → CSS selector-based search.

Example:-

```
soup.find('p')
soup.find_all('a')
soup.get_text()
```

```
'Hello'
```

5. Ethical and Legal Considerations

When doing web scraping, you must follow rules:

Ethical considerations:

- Respect website policies (check robots.txt).
- Avoid overloading servers (use delays).
- Do not scrape sensitive/private data.

Legal considerations:

- Follow terms of service of websites.
- Avoid copyrighted or restricted content.
- Ensure data is used responsibly.

```
import requests
from bs4 import BeautifulSoup
# Step 1: Define the URL
url = "https://example.com"
# Step 2: Send HTTP request
response = requests.get(url)
# Step 3: Check if request was successful
if response.status_code == 200:
    print("Successfully fetched the webpage!\n")
    # Step 4: Parse HTML content
    soup = BeautifulSoup(response.text, "html.parser")
    # Step 5: Extract all links (anchor tags)
    links = soup.find_all('a')
    print("Links found on the webpage:\n")
    for link in links:
        text = link.get_text()
        href = link.get('href')
        print("Text:", text)
        print("Link:", href)
        print("-" * 40)
else:
    print("Failed to retrieve webpage. Status code:", response.status_code)
```

```
Successfully fetched the webpage!

Links found on the webpage:

Text: Learn more
Link: https://iana.org/domains/example
----------------------------------------
```

Conclusion:-

Web scraping using tools like Requests and BeautifulSoup enables efficient data extraction, but must always be done responsibly, ethically, and legally. This practical demonstrated how to programmatically fetch web content and parse it to extract specific information, such as links. Web scraping is a powerful technique for data collection from the internet, vital for tasks like market research, data analysis, and content aggregation, provided it adheres to legal frameworks and website policies.